

Rishi Shah

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EDUCATION

Cornell University

Bachelor of Science in Computer Science (3.7 GPA)

Ithaca, NY

Aug. 2023 – May 2027

- Coursework: Operating Systems, Networks, Data Structures and Algorithms, Databases, Machine Learning, Backend Development, Computer Systems, Functional and Object-Oriented Programming

EXPERIENCE

Microsoft

Incoming SWE Intern, Azure Core

Aug. 2026 – Dec. 2026

Redmond, WA

Apple

SWE Intern, Apple Pay

May 2026 – Aug. 2026

New York City, NY

- Designed a reconciliation and republishing engine (Kotlin/Spring Boot, gRPC) to coordinate on-device deanonymization of payment events, tracking return rates per batch and republishing until analytics thresholds are met.
- Leveraged distributed concurrency controls (Shedlock, mutex) and crash-safe operation ordering across pod failure scenarios to deliver 99.99% reliability at 2,500 TPS sustained, shipping as part of iOS 27.
- Secured distributed infrastructure (DynamoDB, Kafka, Spring Boot on AWS) against gzip bombs, DDoS, malicious payloads, and replay attacks; owned Datadog/Splunk alerting thresholds and observability setup end-to-end.

Capital One

SWE Intern

June 2025 – Aug. 2025

McLean, VA

- Migrated finance ETL from legacy Apache to serverless AWS (Fargate, Batch, Lambda, Step Functions); cut runtime from 8 hrs to 5 min across 500K monthly transactions at 5 companies. First AWS CDK adoption org-wide.
- Built full-stack mentor-matching app using ML vector embeddings (React, TypeScript, Next.js, FastAPI, PostgreSQL, Scikit-learn).

HealthTrends.AI

Founding Machine Learning Engineer

May 2024 – Aug. 2024

New York City, NY

- Built time series models (ARIMA, Prophet, XGBoost) to forecast employee absenteeism; achieved 60% one-month-ahead accuracy, evaluated via MAE, R^2 , and AIC. Presented findings to C-suite to inform product direction.

PROJECTS

Pixel (1st at Google DeepMind × LangChain Multimodal Hacks) | *LangChain, Pinecone, DeepGaze, Gemini*

- Built a multi-agent platform for ad creative optimization: used DeepGaze to model human visual attention and generate scanpath overlays, then applied targeted edits based on predicted attention lift.
- Orchestrated competitor analysis agents over a Pinecone RAG index with LangChain; scored Gemini-generated variants by attention delta against the original creative.

WaveFinder — RAG Music Recommendation System | *ChromaDB, Gemini, TF-IDF, Spotify API*

- Built a music recommendation engine with hybrid retrieval: semantic vector search (ChromaDB) for mood/vibe queries and TF-IDF + popularity-weighted ranking for chart-based queries, routed via a heuristic classifier.
- Grounded results in live Spotify catalog data; generated natural-language summaries per result set using Gemini.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, C, JavaScript, TypeScript, OCaml, SQL

Frameworks: Springboot, MERN (JS full-stack), Flask, postgresSQL, Scikit-learn, Pandas

Tools: AWS (DynamoDB, Lambda, Step Functions, Fargate, S3, Batch), Azure, Apache (Kafka, Iceberg), Datadog, LangChain, Pinecone, Splunk, CI/CD, Docker, Git/GitOps, Gradle, Firebase

ON-CAMPUS

Teaching Assistant — CS 4852: Networks II (Market Design)

Jan. 2024 – Jan. 2026

Resident Advisor

Jan. 2024 – Jan. 2026

ACHIEVEMENTS

Nat'l Merit, CAPTRUST, H&R Block Scholars. NYTW Multimodal, Cornell Fintech Hacks Winners. AIME qualifier.